

End-of-Unit Test — Elec-S1 Part 1

Foundations of DC circuit theory · sections 1.A to 3.C

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 45 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 50 marks.**

Exercise 1. Unit conversions [1.A · 1.B · 2.C]

(3 marks)

Give each result exactly, in scientific notation, without rounding.

- (a) $u = 0.047 \text{ kV}$ in millivolts. (b) A battery of capacity $Q = 3000 \text{ mA h}$, in coulombs. (c) $S = 1.5 \text{ mm}^2$ in square metres.

(a) $u = 0.047 \times 10^3 \text{ V} = 47 \text{ V} = 47 \times 10^3 \text{ mV}$	$u = 4.7 \times 10^4 \text{ mV}$
(b) $Q = 3000 \times 10^{-3} \text{ A} \times 3600 \text{ s} = 3.000 \times 3600 \text{ A s}$	$Q = 1.08 \times 10^4 \text{ C}$
(c) $S = 1.5 \times (10^{-3} \text{ m})^2$	$S = 1.5 \times 10^{-6} \text{ m}^2$

Marking

①	0	1
②	0	1
③	0	1

Marking scheme

- ① 1 pt $4.7 \times 10^4 \text{ mV}$.
 ② 1 pt $1.08 \times 10^4 \text{ C}$.
 ③ 1 pt $1.5 \times 10^{-6} \text{ m}^2$.

Exercise 2. Electric current — definition [1.B]

(5 marks)

Define the electric current i through a cross-section of a conductor. Write down the relation between i , the charge dq that crosses the section and the duration dt . State the name and the symbol of the SI unit of each of these three quantities, and say which of these units are SI base units.

The electric current is the **flow rate of electric charge** through the section: the charge that crosses the section per unit time. Its conventional direction is that of the positive charges (opposite to the motion of the electrons in a metal).

$$i = \frac{dq}{dt}$$

i : **ampere (A)**; q : **coulomb (C)**; t : **second (s)**.

The ampere and the second are SI base units. The coulomb is a derived unit:
 $1 \text{ C} = 1 \text{ A s}$.

Marking

①		
0	1	2
☐	☐	☐

②		
0	1	2
☐	☐	☐

③	
0	1
☐	☐

Marking scheme

- ① **2 pts** Charge crossing the section per unit time (1 pt only for “a flow of electrons” with no “per unit time”).
 ② **2 pts** $i = dq/dt$.
 ③ **1 pt** A, C and s with their exact symbols; ampere (and second) base, coulomb derived.

Examiner's tip

This is Test 1, Exercise 2, in which the unit mark was lost (“C s⁻¹” for the current, “coulombs (s)” for the charge). The answer expected was ampere (A), coulomb (C), second (s).

Exercise 3. Measuring instruments [1.A · 1.B]

(3 marks)

A voltmeter has its V terminal on B and its COM terminal on A. It displays +3.0 V. Therefore:

- ☐ $u_{AB} = 3.0 \text{ V}$;
☒ $u_{AB} = -3.0 \text{ V}$;
☐ $V_A = 3.0 \text{ V}$;
☐ the voltmeter is wrongly connected, so nothing can be concluded.

Why: the display is $V_V - V_{\text{COM}} = V_B - V_A = u_{BA}$, so $u_{AB} = -u_{BA} = -3.0 \text{ V}$.

To measure the current through a resistor R , an ammeter of internal resistance R_A is connected:

- ☐ in parallel across R , with $R_A \gg R$ ☒ in series with R , with $R_A \ll R$
☐ in series with R , with $R_A \gg R$ ☐ in parallel across R , with $R_A \ll R$

Why: the current must flow through the ammeter, and R_A must not reduce it.

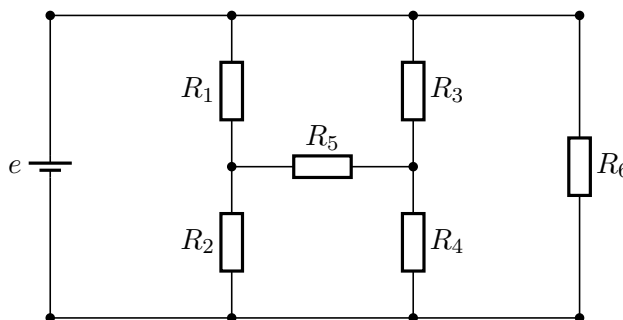
A voltmeter of internal resistance R_V is connected across a resistor R . The measurement is not distorted provided that:

- ☐ $R_V \ll R$ ☐ $R_V = R$ ☒ $R_V \gg R$ ☐ $R_V = 0$

Why: otherwise the voltmeter diverts part of the current flowing through R .

Exercise 4. Reading a circuit diagram [1.C · 3.A]

(4 marks)



Base your answers on the diagram exactly as drawn.

How many nodes?

- ☐ 2
 ☒ 4
 ☐ 6
 ☐ 10

Why: the whole top wire is one node (e , R_1 , R_3 , R_6), the whole bottom wire is another (e , R_2 , R_4 , R_6), and the two middle points are nodes (R_1 , R_2 , R_5 and R_3 , R_4 , R_5). “6” counts the drawn junction dots.

How many branches?

- ☐ 5
 ☐ 6
 ☒ 7
 ☐ 9

Why: each of the 7 components joins two different nodes, so each one is a branch.

How many independent node equations can be written?

- ☒ 3
 ☐ 4
 ☐ 6
 ☐ 7

Why: $N - 1 = 4 - 1 = 3$: the equation of the last node follows from the others.

How many independent meshes?

- ☐ 3
 ☒ 4
 ☐ 5
 ☐ 7

Why: $m = B - N + 1 = 7 - 4 + 1 = 4$ (the four windows of the diagram).

Exercise 5. Characteristics and Ohm’s law [2.A · 2.B]

(3 marks)

The characteristic $u = g(i)$ of an ohmic conductor is a straight line through the origin, of slope 2.0 V mA^{-1} . Its conductance is:

- ☐ 2.0 mS
 ☒ 0.50 mS
 ☐ 2.0 kS
 ☐ 500 S

Why: the slope of $u = g(i)$ is $R = 2.0 \text{ V}/1 \text{ mA} = 2.0 \text{ k}\Omega$, so $G = 1/R = 0.50 \text{ mS}$.

In the generator convention, Ohm’s law for a resistor reads:

- ☐ $u = Ri$
☒ $u = -Ri$
☐ $u = i/R$
☐ $u = -i/R$

Why: in the generator convention the arrows of u and i point the same way, so the sign changes.

In the (u, i) plane, a vertical line $u = E$ is the characteristic of:

- ☒ an ideal voltage source
 ☐ an ideal current source
☐ an ideal diode
 ☐ an ohmic conductor

Why: the voltage is E whatever the current.

Exercise 6. Drift velocity and resistance of a cable [1.B · 2.B · 2.C]

(7 marks)

A copper cable of length L and cross-sectional area S carries a steady current i .

(a) Determine the drift speed v of the conduction electrons. (b) How long does an electron take to travel the whole length of the cable? Why does a lamp at the end of the cable nevertheless light up almost instantly? (c) Determine the resistance R of the cable, then the power p it dissipates.

$L = 10 \text{ m}$	$S = 2.5 \text{ mm}^2$	$i = 4.0 \text{ A}$
$n = 8.0 \times 10^{28} \text{ m}^{-3}$	$e = 1.6 \times 10^{-19} \text{ C}$	$\rho = 1.7 \times 10^{-8} \Omega \text{ m}$

(a) $i = nevS \Rightarrow v = \frac{i}{neS}$ with $S = 2.5 \times 10^{-6} \text{ m}^2$

$$v = \frac{4.0 \text{ A}}{8.0 \times 10^{28} \text{ m}^{-3} \times 1.6 \times 10^{-19} \text{ C} \times 2.5 \times 10^{-6} \text{ m}^2}$$

$$= \frac{4.0}{8.0 \times 1.6 \times 2.5 \times 10^{28-19-6}} \text{ m s}^{-1} = \frac{4.0}{3.2 \times 10^4} \text{ m s}^{-1}$$

$$v = 1.25 \times 10^{-4} \text{ m s}^{-1} \approx 1.3 \times 10^{-4} \text{ m s}^{-1}$$

(b) $\Delta t = \frac{L}{v} = \frac{10 \text{ m}}{1.25 \times 10^{-4} \text{ m s}^{-1}} \quad \Delta t = 8.0 \times 10^4 \text{ s} \approx 22 \text{ h}$

The lamp lights at once because it is the *electric field* (the signal) that is set up along the whole circuit almost at the speed of light. All the electrons, including those already in the lamp, start drifting together. The drift velocity is not the speed of the signal.

(c) $R = \rho \frac{L}{S} = \frac{1.7 \times 10^{-8} \times 10}{2.5 \times 10^{-6}} \Omega = 6.8 \times 10^{-2} \Omega \quad R = 68 \text{ m}\Omega$

$$p = Ri^2 = 6.8 \times 10^{-2} \Omega \times (4.0 \text{ A})^2 = 6.8 \times 10^{-2} \times 16 \text{ W} \quad p \approx 1.1 \text{ W}$$

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1
⑥	0	1
⑦	0	1

Marking scheme

- ① 1 pt Literal expression $v = i/(neS)$, letters only.
- ② 1 pt S in m^2 and numerical expression written with units.
- ③ 1 pt $v = 1.3 \times 10^{-4} \text{ m s}^{-1}$ (the power of ten is the point).
- ④ 1 pt $\Delta t = L/v = 8.0 \times 10^4 \text{ s} \approx 22 \text{ h}$.
- ⑤ 1 pt Signal (field) versus drift of the carriers.
- ⑥ 1 pt $R = \rho L/S = 68 \text{ m}\Omega$.
- ⑦ 1 pt $p = Ri^2 \approx 1.1 \text{ W}$.

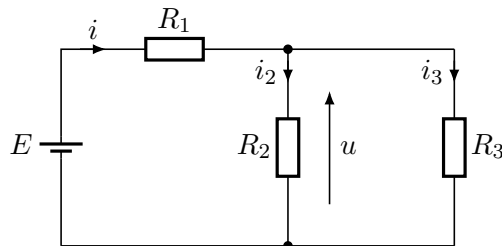
Examiner's tip

Test 1, Ex. 4 revisited: letters first ($v = i/neS$), then numbers *with units*, then the powers of ten on their own: $8.0 \times 1.6 \times 2.5 = 32$ and $10^{28-19-6} = 10^3$.

Exercise 8. A resistive circuit [3.A]

(6 marks)

Determine the equivalent resistance R_{eq} seen by the source, the current i it delivers, the voltage u across R_2 (use a voltage divider), then the currents i_2 and i_3 . Check the node law.



$E = 12 \text{ V}$	$R_1 = 2.0 \text{ k}\Omega$	$R_2 = 6.0 \text{ k}\Omega$	$R_3 = 3.0 \text{ k}\Omega$
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$R_2 \parallel R_3: R_{23} = \frac{R_2 R_3}{R_2 + R_3} = \frac{18}{9.0} \text{ k}\Omega = 2.0 \text{ k}\Omega$				$R_{\text{eq}} = R_1 + R_{23}$
$R_{\text{eq}} = 4.0 \text{ k}\Omega$				
$i = \frac{E}{R_{\text{eq}}} = \frac{12 \text{ V}}{4.0 \text{ k}\Omega}$		$i = 3.0 \text{ mA}$		
Divider (R_1 in series with R_{23}): $u = \frac{R_{23}}{R_1 + R_{23}} E = \frac{2.0}{4.0} \times 12 \text{ V}$				$u = 6.0 \text{ V}$
$i_2 = \frac{u}{R_2} = \frac{6.0 \text{ V}}{6.0 \text{ k}\Omega} = 1.0 \text{ mA}$		$i_3 = \frac{u}{R_3} = \frac{6.0 \text{ V}}{3.0 \text{ k}\Omega} = 2.0 \text{ mA}$		
Node law: $i = i_2 + i_3: 3.0 \text{ mA} = 1.0 \text{ mA} + 2.0 \text{ mA}. \checkmark$				

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1
⑥	0	1

Marking scheme

- ① 1 pt $R_{23} = 2.0 \text{ k}\Omega$.
- ② 1 pt $R_{eq} = 4.0 \text{ k}\Omega$.
- ③ 1 pt $i = 3.0 \text{ mA}$.
- ④ 1 pt $u = 6.0 \text{ V}$ with the divider (not with R_2 alone).
- ⑤ 1 pt $i_2 = 1.0 \text{ mA}$ and $i_3 = 2.0 \text{ mA}$.
- ⑥ 1 pt Node law checked.

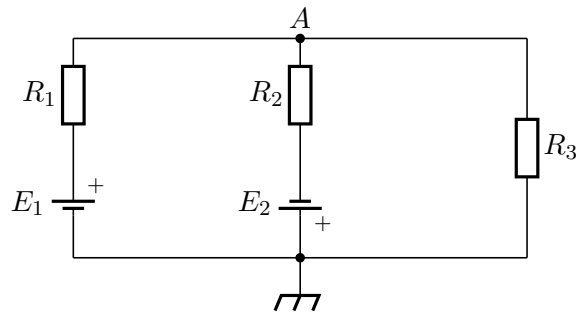
Examiner's tip

The divider formula applies to resistors in *series*: first replace $R_2 \parallel R_3$ by R_{23} , then divide between R_1 and R_{23} .

Exercise 9. Millman's theorem with a reversed source [3.A]

(4 marks)

The ground (bottom wire) is the reference of potentials. Note the orientation of the source E_2 . Determine the potential V_A of node A.



$$E_1 = 12 \text{ V} \quad R_1 = 4.0 \, \Omega \quad E_2 = 4.0 \text{ V} \quad R_2 = 4.0 \, \Omega \quad R_3 = 2.0 \, \Omega$$

The + terminal of E_2 is on the ground side, so the far end of branch 2 is at the potential $-E_2$:

$$V_A = \frac{\frac{E_1}{R_1} + \frac{-E_2}{R_2} + \frac{0}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$\text{Numerator: } \frac{12}{4.0} - \frac{4.0}{4.0} = 3.0 - 1.0 = 2.0 \text{ A}$$

$$\text{Denominator: } \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1.0 \text{ S}$$

$$V_A = 2.0 \text{ V}$$

(between -4 and 12 V ✓)

Marking

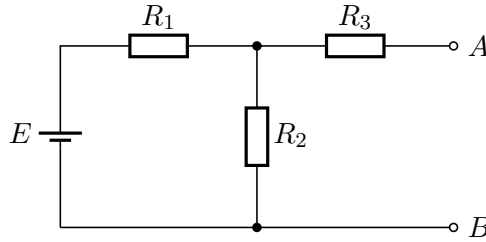
①	0	1
②	0	1
③	0	1
④	0	1

Marking scheme

- ① 1 pt Millman written with the potential $-E_2$ for the reversed branch.
- ② 1 pt Numerator = 2.0 A.
- ③ 1 pt Denominator = 1.0 S.
- ④ 1 pt $V_A = 2.0 \text{ V}$ (with $+E_2$ one finds 4.0 V: 0 for this item).

Exercise 10. Thévenin model and operating point [3.B · 3.C]

(11 marks)



$E = 24 \text{ V}$	$R_1 = 3.0 \Omega$	$R_2 = 6.0 \Omega$	$R_3 = 2.0 \Omega$	$R_c = 12 \Omega$
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- (a) Determine the resistance R_{Th} of the Thévenin model seen from A and B . (b) Determine its electromotive force E_{Th} . (c) Deduce the current I_{No} of the equivalent Norton model.

(a) Sources off (E replaced by a wire): $R_{Th} = R_3 + \frac{R_1 R_2}{R_1 + R_2} = 2.0 + \frac{18}{9.0} = 2.0 + 2.0 \quad \boxed{R_{Th} = 4.0 \Omega}$
(b) Open circuit: no current in R_3 , so u_{AB} is the voltage across R_2 (divider): $E_{Th} = \frac{R_2}{R_1 + R_2} E = \frac{6.0}{9.0} \times 24 \text{ V} \quad \boxed{E_{Th} = 16 \text{ V}}$
(c) $I_{No} = \frac{E_{Th}}{R_{Th}} = \frac{16 \text{ V}}{4.0 \Omega} \quad \boxed{I_{No} = 4.0 \text{ A}}$

Marking

①
0 1
☐ ☐
②
0 1
☐ ☐
③
0 1
☐ ☐
④
0 1
☐ ☐
⑤
0 1
☐ ☐

A load $R_c = 12 \Omega$ is now connected between A and B .

- (d) Determine the operating point (u^*, i^*) analytically. Sketch both characteristics in the (u, i) plane and locate the operating point. (e) Compute the efficiency η of the transfer. (f) Which value of R_c would maximise the power received by the load? Give this maximum power.

(d) $i^* = \frac{E_{Th}}{R_{Th} + R_c} = \frac{16}{4.0 + 12} \text{ A}$ $\boxed{i^* = 1.0 \text{ A}}$ $u^* = R_c i^* \quad \boxed{u^* = 12 \text{ V}}$ Source: line from $(E_{Th}, 0) = (16 \text{ V}, 0)$ to $(0, E_{Th}/R_{Th}) = (0, 4.0 \text{ A})$. Load: line through the origin, slope $1/R_c$. They cross at $(12 \text{ V}, 1.0 \text{ A})$.	
(e) $\eta = \frac{R_c}{R_{Th} + R_c} = \frac{12}{16} \quad \boxed{\eta = 75 \%}$	
(f) Matching: $\boxed{R_c = R_{Th} = 4.0 \Omega} \quad P_c^{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{256}{16} \text{ W}$ $\boxed{P_c^{\max} = 16 \text{ W}} \quad (\text{with } R_c = 12 \Omega: P_c = 12 \text{ W only})$	

Marking

⑥
0 1
☐ ☐
⑦
0 1
☐ ☐
⑧
0 1
☐ ☐
⑨
0 1
☐ ☐
⑩
0 1
☐ ☐
⑪
0 1
☐ ☐

Marking scheme

- ① 1 pt Sources off, $R_{Th} = R_3 + R_1 \parallel R_2$.
- ② 1 pt $R_{Th} = 4.0 \Omega$.
- ③ 1 pt Open circuit, divider across R_2 (no current in R_3).
- ④ 1 pt $E_{Th} = 16 \text{ V}$.
- ⑤ 1 pt $I_{No} = 4.0 \text{ A}$.
- ⑥ 1 pt $i^* = 1.0 \text{ A}$.
- ⑦ 1 pt $u^* = 12 \text{ V}$.
- ⑧ 1 pt Sketch: both lines with their intercepts, intersection marked.
- ⑨ 1 pt $\eta = 75 \%$.
- ⑩ 1 pt $R_c = R_{Th} = 4.0 \Omega$.
- ⑪ 1 pt $P_c^{\max} = 16 \text{ W}$.

Your score: /50 $\longrightarrow$ mark out of 20 = score $\times$ 20/50 = /20

Diagnosis: where did you lose marks?

Ex.	Lecture section(s)	Redo
1	1.A, 1.B, 2.C: prefixes and units	Progress tests 1.A, 1.B, 2.C (Ex. 1)
2	1.B: definition of the current	Progress test 1.B
3	1.A, 1.B: voltmeter and ammeter	Progress tests 1.A (Ex. 3), 1.B (Ex. 3)
4	1.C, 3.A: nodes, branches, meshes	Progress tests 1.C (Ex. 3), 3.A (Ex. 3)
5	2.A, 2.B: characteristics, Ohm	Progress tests 2.A, 2.B
6	1.B, 2.B, 2.C: $i = nevS$, Pouillet, Joule	Progress tests 1.B, 2.C (Ex. 4)
7	2.A: least squares	Progress test 2.A (Ex. 4)
8	3.A: combinations, divider	Progress test 3.A
9	3.A: Millman	Progress test 3.A (Ex. 4)
10	3.B, 3.C: Thévenin, operating point	Progress tests 3.B, 3.C